

Marketing Campaign Analysis — Phase 3: Campaign Performance

Omar Touzani · SQL analysis against marketing_db PostgreSQL (2,212 customers, 23 queries documented)

I used SQL queries against the relational database to measure campaign acceptance rates and cost-efficiency, profile responders vs non-responders, and analyse channel preferences by customer value tier. The source data doesn't include real cost or revenue per campaign, so profitability is framed as a break-even analysis: what revenue per response each campaign needs to be profitable, given its acceptance rate.

Campaign performance and break-even

Campaign	Acceptance rate	Break-even ratio*	Moderate scenario** profit / contact
Latest campaign	15.05%	6.6×	+\$1.51
Campaign 4	7.41%	13.5×	−\$0.78
Campaign 3	7.37%	13.6×	−\$0.79
Campaign 5	7.28%	13.7×	−\$0.82
Campaign 1	6.42%	15.6×	−\$1.07
Campaign 2	1.36%	73.7×	−\$2.59

*Minimum revenue per response ÷ cost per contact needed to break even. Lower = more efficient. **Moderate scenario assumes \$3 cost per contact and \$30 revenue per response.

Who responds to campaigns

Metric	Responders (333)	Non-responders (1,879)	Gap
Avg total spend	\$986	\$540	+82.6%
Avg income	\$60,210	\$50,497	+19.2%
Avg wine spend	\$503	\$270	+86.3%
Avg catalog purchases	4.2	2.4	+75%
Avg past campaigns accepted	0.99	0.18	5.5×
Avg recency (days since last buy)	35.3	51.5	−31.5%

Key findings

The latest campaign is the only profitable one under realistic assumptions. At \$3 cost per contact and \$30 revenue per response, the latest campaign earns \$1.51 per contact. Every historical campaign loses money under the same scenario. Campaign 2 is dramatically inefficient — it would need 73× the cost per contact in revenue per response to break even.

Past campaign acceptance is the single strongest response signal. Responders accepted an average of 0.99 historical campaigns vs 0.18 for non-responders — a 5.5× gap. This is the most discriminating feature in the data and should anchor Phase 4's propensity model.

Catalog is the luxury channel. High spenders use the catalog 20× more than low spenders (5.66 vs 0.29 average purchases). No other channel shows that kind of separation — store and web both scale more gradually with spend. Catalog activity is a stronger marker of high-value customers than any single channel volume metric.

Demographic gaps are real but smaller than behavioural ones. PhD response rate (21.0%) is 5.7× the Basic rate (3.7%). Single and Widow customers respond at 22–24% vs 10–11% for Married/Together. Behavioural features (spend, past campaigns, catalog use) discriminate more sharply than these demographics.

Web channel — high visit volume, low purchase intent per visit. 97.8% of customers buy online at least once, with 5.3 visits per month and only ~4 web purchases over 2 years. The actionable web segment is small: 20 heavy visitors who never buy online — candidates for targeted offers.

What's next

Phase 4 — Customer lifetime value prediction and propensity modelling. Past campaign acceptance, total spend, catalog use, and recency are the top candidate features. *Response* remains the target — *total_hist_campaigns_accepted* is the leakage-safe past-acceptance feature to use.